Do **not** write solutions on this page.

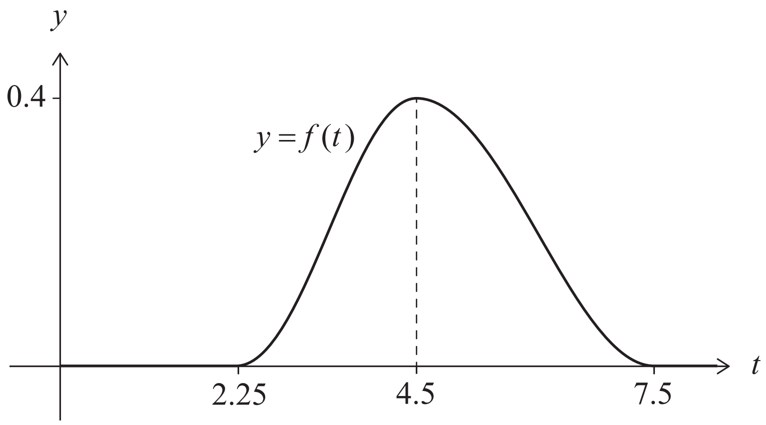
11. [Maximum mark: 19]

In a marathon race, the random variable T represents the time, in hours, taken for a runner to complete the race. No runner completes the race in less than 2.25 hours, and no runner completes it in more than 7.5 hours.

The probability density function for T is modelled by f, defined by

$$f(t) = \begin{cases} \dfrac{4}{21}\left(1 - \cos\left(\dfrac{4\pi}{9}(t - 2.25)\right)\right), & 2.25 \le t < 4.5 \\ \dfrac{4}{21}\left(1 + \cos\left(\dfrac{\pi}{3}(t - 4.5)\right)\right), & 4.5 \le t \le 7.5 \\ 0, & \text{otherwise.} \end{cases}$$

The graph of f has a maximum point at $t = 4.5$ as shown in the following diagram:



(a) (i) Find the value of $\displaystyle\int_{2.25}^{4.5} f(t)\,\mathrm{d}t$.

(ii) Write down the mode of T.

(iii) Determine which is greater, the mode of T or the median of T, justifying your answer. [4]

(This question continues on the following page)

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(Question 11 continued)

The runners who finish the race in 3.5 hours or less are considered to be fast runners.

(b) Find the probability that a runner chosen at random is a fast runner. [2]

(c) Find the probability that a fast runner chosen at random finishes the race in 3 hours or less. [3]

(d) Find the lower quartile of T. [3]

Each runner's time is converted to a score which is calculated as $a - bt$, where t represents their time in hours, and a, $b > 0$.

Consider the random variable P which represents the score of a runner. It is given that $\mathrm{E}(P) = 100$ and the maximum possible score is 150.

(e) Use $\mathrm{E}(T) = 4.723$ to determine the value of a and the value of b, giving your answers to the nearest integer. [5]

(f) Given also that $\mathrm{Var}(T) = 0.906$, find $\mathrm{Var}(P)$. [2]